Supplementary material for "Of Cats and Dogs: What Languages Do Mitochondrions Speak? A Vector-Space Analysis of mtDNA Vocabulary"

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S1 Maximum-likelihood trees: records, rotation and alignment

Records. One record per species was used. It was the record whose header contains the species name, then the one with the smallest share of characters other than A, C, G and T, then the longest one. The accessions are given in Table S3. *Lu. adusta* was left out because its only record is partial (15,263 bp, 25.5% characters other than A, C, G or T), which leaves 44 Felidae and 28 Canidae species.

Rotation to a common origin. Mitogenomes are circular and the records start at different positions, so every record was rotated to start at the same nucleotide, the first one of the *tRNA-Phe* gene. The start was defined by the first 500 bp of a published record: *Pardofelis marmorata*, `MW257207.1`, for Felidae, and the dog RefSeq record `NC_002008.4` for Canidae. The rotation was found by matching unique 12-mers between each record and this anchor, with the strand checked at the same time. All records had a clear match and none had to be reverse-complemented. In Canidae, 22 of the 28 records already started at this position (offset 0), and the largest offsets are those of the records that start at another gene, such as the *COX1* start of *V. cana*, *V. macrotis*, and *V. velox* (about 11,200–11,300 bp). In Felidae, only 8 of the 44 records needed no rotation (two more differ by 1 bp), because many Felidae records start in the control region; most offsets are between 100 and 950 bp (Table S3).

Outgroups. The spotted hyena (*Crocuta crocuta*, `NC_020669.1`) was added to the Felidae alignment and the brown bear (*Ursus arctos*, `NC_003427.1`) to the Canidae alignment. The outgroup was used to root the tree and removed afterwards.

Alignment and tree. The sequences were aligned with MAFFT v7.505 (option `--auto`, which chose FFT-NS-2), and the trees were estimated with IQ-TREE 2.0.7 under the GTR+F+Γ_4 model (empirical base frequencies, four rate categories; $\alpha = 0.21$), with 1,000 ultrafast bootstrap replicates. The Felidae alignment has 19,398 columns for 45 sequences (44 species and the outgroup), of which 12.8% are gaps or ambiguity codes; the Canidae alignment has 17,380 columns for 29 sequences. The model was not chosen by model selection, and the control region was kept in the alignment. Ten of the 45 sequences (nine Felidae and the outgroup) failed the test of equal base composition of IQ-TREE, which the model assumes.

S2 K2P distances and the origin of the records

The K2P implementation (main text, Section 4.5) was not corrected for the origin of the circular sequences or for ambiguous characters, and the Canidae distance matrix shows the consequences. The distances are 0.42–0.50 for the record with 25% ambiguous characters (*Lu. adusta*), and 0.48–0.65 between the *C. latrans*, *C. lycaon* and *C. rufus* records (15,458 bp, MZ36789x series) and the *V. cana*, *V. macrotis* and *V. velox* records, which start at the *COX1* gene (`PZ823142`–`PZ823144`), against 0.16–0.19 for other pairs. The unusual K2P root of the Canidae tree (the ten *Vulpes* species against all others) may be related to the three *Vulpes* records that start at *COX1*; this was not tested. Four records have more than 5% ambiguous characters (`KT448271.1` and three *C. familiaris* records; Appendix C), and all records were kept.

S3 Clades of the reference recovered by the UPGMA trees

Table S4 gives, for each tree, the number of clades of the pruned reference topology that the tree contains. The sections below describe which clades these are.

S3.1 Felidae

All eight genera with more than one species (*Felis*, *Leopardus*, *Lynx*, *Panthera*, *Prionailurus*, *Caracal*, *Catopuma*, *Neofelis*) are monophyletic in every tree.

Clades recovered by all six trees (18). Four clades join several genera: *Puma* with *Herpailurus*, the puma lineage with *Acinonyx*, the caracal lineage (*Caracal* and *Leptailurus*), and the 37 species other than *Panthera* and *Neofelis*. The other 14 are the genera *Leopardus* (11 species), *Felis* (7), *Panthera* (5), *Prionailurus* (5) and *Lynx* (4), and nine smaller groups inside *Felis*, *Lynx*, *Prionailurus*, *Caracal*, *Catopuma*, *Neofelis* and *Panthera*.

Clades recovered by no tree (17). Three deep clades of 34, 23 and 16 species; the clade of the bay cat lineage and *Lynx* (seven species); the clade of *Catopuma* and *Pardofelis marmorata*; the clade of *Otocolobus* and *Prionailurus*; and 11 clades inside genera: eight in *Leopardus* (clades of 7, 4, 4, 3, 2, 2, 2, and 2 species), and one each in *Felis* (six species), *Lynx* (*L. lynx* with *L. pardinus*), and *Panthera* (the tiger with the snow leopard). The deeper relationships are therefore not recovered by any of the distances, which is expected for a single mitochondrial locus and UPGMA.

Differences among the distances. The TF-IDF cosine tree recovers every clade that the species-level cosine and the species-level Euclidean trees recover, and five and four clades more, respectively. Compared with the Jaccard tree, the K2P tree and the genome-level Euclidean tree, it gains three clades and loses one (*Otocolobus* with *Prionailurus* and *Felis*). The wildcat group of *Felis* (*F. silvestris*, *F. lybica*, *F. catus*, *F. bieti*) is recovered by the Jaccard, K2P, genome-level Euclidean and TF-IDF trees. Only the TF-IDF tree and the genome-level Euclidean tree join *Neofelis* with *Panthera*; the other four trees place *Neofelis* outside *Panthera* and the remaining genera. The three trees with RF 40/84 (Jaccard, K2P, genome-level Euclidean) are not identical.

Within genera. No tree joins the snow leopard next to the tiger; five trees place it with the lion and the leopard (the TF-IDF tree also with the jaguar). All six trees join *Lynx canadensis* with *Lynx lynx*, whereas the reference joins *Lynx lynx* with *Lynx pardinus*. In *Leopardus*, *L. colocola* falls with *L. tigrinus* and *L. emiliae*, and *L. tilcayo* is the sister of *L. geoffroyi*, with *L. guigna* next to them, in all six trees; the reference places *L. tilcayo* with *L. pardinoides* and *L. tigrinus*.

Genera collapsed. With genera collapsed to single leaves (14 leaves, maximum 24), the distances to the reference are 12 for the genome-level Euclidean distance, 14 for the Jaccard distance, K2P and the TF-IDF cosine, and 16 for both species-level distances.

S3.2 Canidae

29 species (27 reference clades). Nine clades are recovered by all six trees: the genera *Canis* (eight species) and *Vulpes* (ten species), the group of the coyote, the red wolf and the eastern wolf, the pair of the wolf and the dog, the pair of *Urocyon* species, and four groups inside *Vulpes* (*V. lagopus*, *V. macrotis*, and *V. velox*; *V. macrotis* with *V. velox*; *V. corsac* with *V. ferrilata*; *V. rueppellii* with *V. vulpes*). Ten clades are recovered by no tree: the root of the reference (*Urocyon* against the other 27 species); the clades of *Canis*, *Cuon* and *Lycaon* (ten species), and of *Canis* and *Cuon* (nine); the pair of *Lupulella* species; the pair *V. cana* with *V. zerda*; three groupings inside *Vulpes* (of 8, 7, and 4 species); and two groupings inside *Canis* (of 7 and 5 species). The 5-species one

is the coyote group with the wolf and the dog; it is missed because *C. lupaster* is joined with the wolf and the dog. The genera *Canis*, *Vulpes*, and *Urocyon* are monophyletic in every tree, and the two species of *Lupulella* are separated in all six trees.

Canis. The wolf and the dog are joined with *C. lupaster* in all six trees, whereas in the reference *C. lupaster* lies outside the group formed by the wolf, the dog, and the coyote group. The red wolf and the eastern wolf form a pair in the TF-IDF, species-level cosine, and Euclidean trees, as in the reference; in the Jaccard, K2P, and genome-level Euclidean trees, the red wolf is the sister of the coyote and the eastern wolf.

Differences among the distances. The Jaccard distance and the species-level cosine recover 14 clades each, but the two trees differ. The TF-IDF tree has three clades more and four fewer than the Jaccard tree, and one clade more and two fewer than the species-level cosine tree.

Roots. The Jaccard, TF-IDF, and species-level cosine trees split the 29 species into two groups of 15 and 14 (the South American canids, the wolf-like canids, and *Lycaon* against the foxes and *Urocyon*). K2P, the genome-level Euclidean distance, and the species-level Euclidean distance place *Lu. adusta* alone at the root, which is the effect of its partial record (S1).

28 species (26 reference clades), without *Lu. adusta*. All genera are monophyletic in every tree. Ten reference clades are recovered by all six trees and nine by none. All trees but K2P have two groups of 14 at the root; the K2P tree has the ten *Vulpes* species against all others.

S4 Newick strings

The trees estimated from the data are given in the supplementary files `https://github.com/andrijrovenchak/cats_dogs`, one tree per file in Newick format. The UPGMA trees are named `<family>_<distance>_tree.nwk`, where `<family>` is `felidae` (44 species), `canidae29` (29 species), or `canidae28` (28 species, without *Lu. adusta*), and `<distance>` is `jaccard`, `mash`, `k2p`, `euclidean`, `tfidf`, `cosine-species` or `euclidean-species`, which gives 21 trees. The maximum-likelihood trees (`felidae_mt_ml.nwk` and `canidae28_mt_ml.nwk`) are rooted on the outgroup with the outgroup removed. The same trees with the outgroup, branch lengths and ultrafast bootstrap values are in `felidae_mt_ml_support.nwk` and `canidae28_mt_ml_support.nwk`. The reference topologies, full (48 and 39 leaves) and pruned to the species of the collection, are in `reference_trees.nwk`.

The full reference trees are as follows:

Felidae. ((((Panthera_tigris,Panthera_uncia),(Panthera_onca,(Panthera_leo,
Panthera_pardus))),(Neofelis_nebulosa,Neofelis_diardi)),(((Caracal_caracal,
Caracal_aurata),Leptailurus_serval),((((Leopardus_pardalis,(Leopardus_wiedii,
Leopardus_jacobita)),(Leopardus_braccatus,Leopardus_colocola,Leopardus_garleppi,
Leopardus_fasciatus,Leopardus_pajeros)),(((Leopardus_guttulus,Leopardus_emiliae),(
Leopardus_geoffroyi,Leopardus_guigna)),(Leopardus_tilcayo,(Leopardus_pardinoides,
Leopardus_tigrinus)))),((((Catopuma_temminckii,Catopuma_badia),
Pardofelis_marmorata),(Lynx_rufus,(Lynx_canadensis,(Lynx_lynx,Lynx_pardinus)))),((
Acinonyx_jubatus,(Puma_concolor,Herpailurus_yagouaroundi)),((Otocolobus_manul,(
Prionailurus_rubiginosus,(Prionailurus_planiceps,(Prionailurus_viverrinus,(
Prionailurus_bengalensis,Prionailurus_javanensis))))),(Felis_chaus,(
Felis_nigripes,(Felis_margarita,(Felis_silvestris,(Felis_bieti,(Felis_lybica,
Felis_catus)))))))))));

Canidae. ((Urocyon_cinereoargenteus,Urocyon_littoralis),((Otocyon_megalotis,((
Nyctereutes_procyonoides,Nyctereutes_viverrinus),((Vulpes_zerda,Vulpes_cana),(
Vulpes_chama,((((Vulpes_vulpes,(Vulpes_rueppellii,Vulpes_pallida)),
Vulpes_bengalensis),(Vulpes_corsac,Vulpes_ferrilata)),(Vulpes_lagopus,(
Vulpes_macrotis,Vulpes_velox)))))),(((Speothos_venaticus,Chrysocyon_brachyurus),(
Atelocynus_microtis,(Cerdocyon_thous,(Lycalopex_vetula,(Lycalopex_sechurae,((
Lycalopex_fulvipes,Lycalopex_gymnocerca),(Lycalopex_grisea,
Lycalopex_culpaeus)))))),((Lupulella_adusta,Lupulella_mesomelas),(Lycaon_pictus,(
Cuon_alpinus,(Canis_aureus,(Canis_simensis,(Canis_lupaster,((Canis_latrans,(
Canis_rufus,Canis_lycaon)),(Canis_lupus,Canis_familiaris)))))))))));

Supplementary tables

Table S1: Records of the collections and the records sampled for the analyses (at most three per species). *Both*: records in the sample for the species documents (TF-IDF, cosine, Euclidean; classification) and in the sample for the pairwise distances (Jaccard, Mash, K2P, Euclidean between genomes). *Documents only* and *distances only*: records found in one sample only. "all" means that the species has at most three records, which are all used. The complete list of the 2,056 records (578 Felidae, 1,478 Canidae) with lengths and the share of characters other than A, C, G and T is given in the file `accessions_all.csv`.

Species	Records	Both	Documents only	Distances only
Felidae				
Acinonyx jubatus	4	AY463959.1, KP202271.1, NC_005212.1	–	–
Caracal aurata	1	all	–	–
Caracal caracal	5	NC_028306.1, PP566120.1	PP566118.1	PP566117.1
Catopuma badia	3	all	–	–
Catopuma temminckii	42	–	KX224490.1, KX224496.1, KX224529.1	KX224494.1, KX224495.1, KX224523.1
Felis bieti	1	all	–	–
Felis catus	97	–	OR077767.1, OR077779.1, OR085754.1	OR077768.1, OR077830.1, OR085740.1
Felis chaus	1	all	–	–
Felis lybica	1	all	–	–
Felis margarita	2	all	–	–
Felis nigripes	1	all	–	–
Felis silvestris	21	NC_028310.1	PZ499039.1, PZ499060.1	PZ499038.1, PZ499042.1
Herpailurus yagouaroundi	1	all	–	–
Leopardus colocola	1	all	–	–
Leopardus emiliae	1	all	–	–
Leopardus geoffroyi	1	all	–	–
Leopardus guigna	2	all	–	–
Leopardus guttulus	1	all	–	–
Leopardus jacobita	2	all	–	–
Leopardus pardalis	3	all	–	–
Leopardus pardinoides	2	all	–	–
Leopardus tigrinus	2	all	–	–
Leopardus tilcayo	1	all[a)]	–	–
Leopardus wiedii	1	all	–	–
Leptailurus serval	1	all	–	–
Lynx canadensis	1	all	–	–
Lynx lynx	45	–	KR919624.1, MK229210.2, OK539609.1	MK229240.2, MK229287.1, OK539607.1
Lynx pardinus	49	KX911259.1	KX911268.1, KX911375.1	KX911276.1, KX911336.1
Lynx rufus	4	KP202285.1, NC_014456.1, PZ178888.1	–	–

Species	Records	Both	Documents only	Distances only
Neofelis diardi	1	all	–	–
Neofelis nebulosa	4	KP202291.1, OR682198.1	KU133958.1	NC_008450.1
Otocolobus manul	2	all	–	–
Panthera leo	27	–	KC834784.1, KP001493.1, NC_018053.1	KP001500.1, KP001503.1, OK513011.1
Panthera onca	16	OR863186.1	KM236783.1, OR863194.1	NC_022842.1, OR863191.1
Panthera pardus	22	KX655614.1	KJ866876.1, PQ119384.1	KP202265.1, OR817737.1
Panthera tigris	22	BK062824.1	BK062820.1, NC_010642.1	JF357974.1, KF297576.1
Panthera uncia	135	–	MT423712.1, PP646756.1, PP646806.1	MT423717.1, PP646801.1, PP646827.1
Pardofelis marmorata	4	KT288227.2, PV137945.1	MW257207.1	NC_028303.1
Prionailurus bengalensis	13	KP202259.1	KP202258.1, OR282482.1	MN121632.1, NC_016189.1
Prionailurus javanensis	5	KX857760.1, KX857785.1	KX857754.1	KX857769.1
Prionailurus planiceps	16	KY682741.1	KY682728.1, KY682731.1	KY682729.1, KY682734.1
Prionailurus rubiginosus	1	all	–	–
Prionailurus viverrinus	1	all	–	–
Puma concolor	12	KP202261.1, MH818222.1	MH807447.1	MH814704.1
Canidae				
Canis aureus	2	all	–	–
Canis familiaris	1130	EU408260.1, EU789777.1, KU290581.1	–	–
Canis latrans	37	–	DQ480511.1, MZ042351.1, MZ367891.1	MZ042352.1, MZ367887.1, MZ367888.1
Canis lupaster	2	all	–	–
Canis lupus	126	–	AB499821.1, EU789788.1, MZ042334.1	EU789787.1, MZ042339.1, MZ042359.1
Canis lycaon	2	all	–	–
Canis rufus	8	MZ367900.1	MZ367896.1, MZ367912.1	MZ367902.1, MZ367903.1
Canis simensis	1	all	–	–
Chrysocyon brachyurus	1	all	–	–
Cuon alpinus	3	all	–	–
Lupulella adusta	1	all	–	–
Lupulella mesomelas	1	all	–	–
Lycalopex sechurae	1	all	–	–
Lycaon pictus	1	all	–	–
Nyctereutes procyonoides	5	KF709435.1, OM732330.1	MT874472.1	NC_013700.1
Otocyon megalotis	2	all	–	–
Speothos venaticus	1	all	–	–

Species	Records	Both	Documents only	Distances only
Urocyon cinereoargenteus	90	–	KP129087.1, KP129101.1, MW600061.1	MW600001.1, MW600005.1, MW600040.1
Urocyon littoralis	15	KP128924.1	KP128940.1, KP128994.1	KP128989.1, KP128993.1
Vulpes cana	1	all	–	–
Vulpes chama	2	all	–	–
Vulpes corsac	6	PZ198068.1	NC_023958.1, PZ198069.1	PZ198065.1, PZ198067.1
Vulpes ferrilata	1	all	–	–
Vulpes lagopus	5	OR880604.1	KP342451.1, OR880609.1	NC_026529.1, OR880606.1
Vulpes macrotis	1	all	–	–
Vulpes rueppellii	4	NC_087722.1, PZ198071.1	PV802399.1	PZ198070.1
Vulpes velox	1	all	–	–
Vulpes vulpes	26	–	GQ374180.1, KP342452.1, PZ198082.1	PZ198072.1, PZ198075.1, PZ198084.1
Vulpes zerda	2	all	–	–

[a] *L. tilcayo*: mitogenome assembled in this study from the reads of SRR35083483; it has no INSDC accession.

Table S2: Annotation of the recovered *L. tilcayo* mitogenome (17,104 bp, circular; origin arbitrary) by MITOS2. Positions are 1-based and inclusive. Gene names are those of MITOS2: `trn` genes are tRNAs, with the amino acid (one-letter code) after `trn` and the anticodon in brackets; `rrnS` and `rrnL` are the small and large rRNA. The control region is split by the origin of the assembly. The strand is given relative to the sequence in the supplementary file.

Feature	Type	Start	End	Length (bp)	Strand
trnF(gaa)	tRNA	315	384	70	+
rrnS	rRNA	385	1344	960	+
trnV(tac)	tRNA	1343	1410	68	+
rrnL	rRNA	1411	2985	1575	+
trnL2(taa)	tRNA	2987	3061	75	+
nad1	CDS	3064	4020	957	+
trnI(gat)	tRNA	4020	4088	69	+
trnQ(ttg)	tRNA	4086	4159	74	−
trnM(cat)	tRNA	4161	4229	69	+
nad2	CDS	4230	5273	1044	+
trnW(tca)	tRNA	5272	5340	69	+
trnA(tgc)	tRNA	5355	5424	70	−
trnN(gtt)	tRNA	5426	5498	73	−
trnC(gca)	tRNA	5532	5596	65	−
trnY(gta)	tRNA	5597	5663	67	−
cox1	CDS	5665	7209	1545	+
trnS2(tga)	tRNA	7207	7275	69	−
trnD(gtc)	tRNA	7282	7350	69	+
cox2	CDS	7351	8034	684	+
trnK(ttt)	tRNA	8038	8105	68	+
atp8	CDS	8107	8310	204	+
atp6	CDS	8268	8948	681	+
cox3	CDS	8948	9732	785	+
trnG(tcc)	tRNA	9732	9800	69	+
nad3	CDS	9807	10157	351	+
trnR(tcg)	tRNA	10148	10216	69	+
nad4l	CDS	10217	10513	297	+
nad4	CDS	10507	11884	1378	+
trnH(gtg)	tRNA	11885	11953	69	+
trnS1(gct)	tRNA	11954	12012	59	+
trnL1(tag)	tRNA	12013	12082	70	+
nad5	CDS	12074	13903	1830	+
nad6	CDS	13887	14411	525	−
trnE(ttc)	tRNA	14415	14483	69	−
cob	CDS	14487	15626	1140	+
trnT(tgt)	tRNA	15627	15696	70	+
trnP(tgg)	tRNA	15697	15763	67	−
control_region	D-loop	15764	17104	1341	–
control_region	D-loop	1	314	314	–

Table S3: Records used for the maximum-likelihood trees (one per species) and their rotation to a common origin, the first nucleotide of the *tRNA-Phe* gene. Length is in bp; non-ACGT is the share of characters other than A, C, G and T in percent; offset is the rotation applied to the circular record (in bp, modulo the length); votes is the number of 12-mers shared with the anchor record at that offset; identity is the share of identical positions in the first 120 bp after rotation, compared with the anchor record. All records were on the + strand. The anchor records are `MW257207.1` (Felidae) and `NC_002008.4` (Canidae).

Species	Accession	Length	Non-ACGT	Offset	Votes	Identity
Felidae						
Acinonyx jubatus	`NC_005212.1`	17,047	0	892	274	0.96
Caracal aurata	`NC_028299.1`	16,826	0	764	387	0.98
Caracal caracal	`PP566119.1`	16,903	0	0	322	0.97
Catopuma badia	`NC_028300.1`	16,693	0	711	300	0.97
Catopuma temminckii	`NC_027115.1`	16,985	0	899	310	0.98
Felis bieti	`KP202273.1`	16,809	0	828	305	0.99
Felis catus	`PP070532.1`	17,048	0	904	280	0.99
Felis chaus	`NC_028307.1`	16,653	0	670	304	0.97
Felis lybica	`KP202275.1`	16,735	0	756	297	0.99
Felis margarita	`NC_028308.1`	16,864	0	883	295	0.99
Felis nigripes	`NC_028309.1`	16,723	0	760	272	0.98
Felis silvestris	`OR095103.1`	16,437	0	573	309	0.99
Herpailurus yagouaroundi	`NC_028311.1`	16,700	0	698	290	0.85
Leopardus colocola	`NC_028314.1`	16,650	0	752	353	1.00
Leopardus emiliae	`OR822034.1`	17,581	0	382	341	1.00
Leopardus geoffroyi	`NC_028320.1`	16,738	0	772	285	0.97
Leopardus guigna	`OR822031.1`	17,417	0	195	276	0.97
Leopardus guttulus	`NC_085265.1`	17,944	0	194	314	0.96
Leopardus jacobita	`OR822033.1`	17,290	0	103	283	0.97
Leopardus pardalis	`OR822032.1`	17,245	0	119	335	0.97
Leopardus pardinoides	`OR257582.1`	17,419	0	126	307	0.97
Leopardus tigrinus	`NC_028317.1`	16,767	0	707	319	0.97
Leopardus tilcayo	SRR35083483[a]	17,104	0	314	276	0.97
Leopardus wiedii	`NC_028318.1`	16,704	0	727	311	0.97
Leptailurus serval	`NC_028316.1`	16,708	0	651	330	0.97
Lynx canadensis	`NC_028313.1`	16,666	0	665	343	0.98
Lynx lynx	`KR919624.1`	17,054	0	0	354	0.98
Lynx pardinus	`NC_028319.1`	16,806	0	805	302	0.97
Lynx rufus	`PV369935.1`	17,114	0	0	347	0.98
Neofelis diardi	`NC_072125.1`	16,094	0.01	16,093	77	0.82
Neofelis nebulosa	`KU133958.1`	16,859	0	907	265	0.82
Otocolobus manul	`MH978908.1`	16,672	0	655	241	0.97
Panthera leo	`KP001506.1`	17,128	0	953	258	0.86
Panthera onca	`PX668385.1`	17,085	0	0	232	0.93
Panthera pardus	`KP001507.1`	17,091	0	951	223	0.83
Panthera tigris	`MH124079.1`	17,115	0	1	202	0.82
Panthera uncia	`PZ790875.1`	16,976	0	0	275	0.95
Pardofelis marmorata	`MW257207.1`	17,075	0	0	501	1.00
Prionailurus bengalensis	`OR282482.1`	17,246	0	0	318	0.98
Prionailurus javanensis	`KX857785.1`	15,443	0.04	0	313	0.98
Prionailurus planiceps	`NC_028312.1`	16,709	0	728	326	0.98
Prionailurus rubiginosus	`NC_028304.1`	16,832	0	853	311	0.97
Prionailurus viverrinus	`NC_028305.1`	16,723	0	752	295	0.98
Puma concolor	`NC_016470.1`	17,153	0	914	324	0.98
Crocuta crocuta[†]	`NC_020669.1`	17,112	0	786	153	0.95

Species	Accession	Length	Non-ACGT	Offset	Votes	Identity
Canidae						
Canis aureus	NC_067757.1	16,669	0	0	383	0.98
Canis familiaris	AY729880.1	16,756	0	0	501	1.00
Canis latrans	MZ042353.1	16,725	0	0	396	0.98
Canis lupaster	KT378606.1	16,734	0	0	437	0.99
Canis lupus	PZ198063.1	16,824	0	0	489	1.00
Canis lycaon	MZ367897.1	15,458	0.01	0	384	0.98
Canis rufus	NC_062617.1	16,474	0	342	383	0.98
Canis simensis	NC_062616.1	16,676	0.02	483	394	0.99
Chrysocyon brachyurus	NC_024172.1	16,760	0	0	322	0.94
Cuon alpinus	KF646248.1	16,767	0	0	311	0.95
Lupulella mesomelas	KT448280.1	16,069	0.35	0	342	0.97
Lycalopex sechurae	KT448284.1	16,068	0.34	0	300	0.97
Lycaon pictus	NC_028427.1	16,173	0	0	292	0.95
Nyctereutes procyonoides	KF709435.1	16,802	0	0	259	0.96
Otocyon megalotis	MW257223.1	16,753	0.01	0	220	0.93
Speothos venaticus	NC_053974.1	16,975	0.09	16,972	278	0.85
Urocyon cinereoargenteus	MW600040.1	16,475	0	0	207	0.86
Urocyon littoralis	KP128924.1	16,469	0	0	195	0.86
Vulpes cana	PZ823142.1	16,578	0	11,229	183	0.91
Vulpes chama	ON756054.3	16,772	0	0	215	0.92
Vulpes corsac	PZ198066.1	16,831	0	0	263	0.93
Vulpes ferrilata	NC_027935.1	16,667	0	0	298	0.94
Vulpes lagopus	OR880606.1	16,657	0	0	293	0.93
Vulpes macrotis	PZ823143.1	16,617	0	11,267	281	0.93
Vulpes rueppellii	PZ198071.1	16,751	0	0	239	0.93
Vulpes velox	PZ823144.1	16,655	0	11,305	281	0.93
Vulpes vulpes	PZ198081.1	16,821	0	0	246	0.93
Vulpes zerda	KJ603240.1	16,734	0	0	223	0.93
Ursus arctos[†]	NC_003427.1	17,020	0	1,137	78	0.73

[a] *L. tilcayo*: mitogenome assembled in this study from the reads of SRR35083483; it has no INSDC accession. [†] Outgroup (*Crocuta crocuta* for Felidae, *Ursus arctos* for Canidae), removed from the tree after rooting.

Table S4: Number of reference clades found in the UPGMA trees (Table 2 gives the corresponding RF distances).

	Felidae	Canidae	Canidae*
Reference clades	42	27	26
Jaccard (13-mers)	22	14	14
K2P (pairwise alignment)	22	10	12
Euclidean (genomes)	22	10	12
TF-IDF cosine (species)	24	13	13
Cosine (species)	19	14	15
Euclidean (species)	20	11	13
Recovered by all six trees	18	9	10
Recovered by none	17	10	9

* Canidae without *Lu. adusta*.